

# A Tri-Tiered Local LLM Framework for Cost-Efficient Active Learning in Text Classification and Automated Categorization

Saaket Baldawa

CSE(AIML)

Institute of Aeronautical Engineering

Hyderabad, India

24951a66g3@iare.ac.in

Dr. P. Naga Lakshmi Devi

Assistant Professor CSE(AIML)

Institute of Aeronautical Engineering

Hyderabad, India

p.nagalakshmi@iare.ac.in

**Abstract**—Large Language Models (LLMs) offer strong capabilities for text classification, but applying high-capacity models to every instance can increase computational cost, while supervised classification remains dependent on labeled data. This paper presents a tri-tiered local LLM framework for cost-efficient active learning in text classification and automated categorization. The framework combines a lightweight DistilBERT classifier, a locally deployed Qwen2.5 3B model, and a human-annotation tier, with uncertainty-driven routing used to allocate progressively greater resources to difficult instances. Entropy and maximum confidence determine escalation, while the local LLM provides an intermediate decision layer before human intervention is used for extreme-uncertainty cases or model failures. Human-tier labels are accumulated and used to adapt the Tier 1 classifier through an active learning loop. Evaluation on AG News, DBpedia-14, Emotion, and IMDb shows final accuracy gains of 0.4–8.0 percentage points over the Tier 1 classifier in the primary test runs, with simulated human effort ranging from 0.2% to 3.6%. Five-seed evaluations retain positive mean accuracy gains across all four datasets. The results show that uncertainty-based tiering can provide a practical accuracy–resource trade-off, although the benefit and routing behavior vary substantially with dataset difficulty.

**Index Terms**—Active Learning, Human-in-the-Loop Learning, Large Language Models, Local LLMs, Text Categorization, Uncertainty Sampling

## I. INTRODUCTION

Text categorization is a fundamental task in Natural Language Processing (NLP) that assigns textual data to predefined categories and supports applications such as document organization, topic identification, sentiment analysis, and information filtering. As textual data continue to grow in volume and diversity, automated categorization systems must provide accurate predictions while remaining practical to operate at scale. Traditional supervised approaches depend on labeled examples, and obtaining such labels can become a major bottleneck when annotation requires human effort [1], [2].

The development of transformer-based language models and, more recently, Large Language Models (LLMs) has expanded the capabilities of automated text processing. LLMs can perform classification through natural-language instructions and in-context examples, often providing strong semantic

understanding across different tasks [3], [4]. At the same time, using a large model for every input can be computationally expensive, and the strongest model is not necessarily required for every classification decision. This creates a practical tension between predictive capability, inference cost, and the amount of human supervision required [5]–[8].

Active Learning offers a complementary strategy by selectively acquiring labels for informative or uncertain instances rather than labeling an entire unlabeled pool [1], [2]. However, a binary choice between an automated classifier and a human annotator can still be inefficient: some uncertain instances may be resolved by a more capable automated model, while only a smaller subset may genuinely require human judgment. Recent work on LLM-assisted active learning and human annotation similarly explores the use of stronger language models within the annotation and learning process [9]–[13]. This motivates a resource-allocation strategy in which different levels of computational and annotation effort are assigned according to instance uncertainty [5], [6], [8], [14], [15].

In this work, we present a tri-tiered local LLM framework for cost-efficient active learning in text classification and automated categorization. The framework combines a lightweight encoder classifier, a locally deployed LLM, and a human annotation tier. Instances are first processed by the lightweight classifier; uncertain cases are selectively passed to the local LLM, while extreme-uncertainty cases or failed LLM inferences can reach the human tier. Labels obtained through the human tier are accumulated and used to adapt the lightweight classifier. The framework is evaluated across four text-classification datasets under a strict separation of training, active-learning, validation, and test data.

The main contributions of this work are as follows:

- We present a tri-tiered classification architecture that combines a lightweight encoder, a local LLM, and human annotation through uncertainty-driven escalation.
- We implement an entropy- and confidence-based routing policy that selectively invokes higher-cost processing instead of applying a single model to every instance.

- We integrate human-tier labels into an active learning loop that adapts the lightweight classifier using accumulated annotation history.
- We evaluate the framework across four datasets using classification, calibration, human-effort, compute-cost, utility, baseline, ablation, and multi-seed analyses.

## II. RELATED WORK

Large Language Models have increasingly been investigated for text classification because they can perform classification through natural-language instructions and in-context examples. Kostina et al. compared several LLMs with machine-learning and transformer-based classifiers and showed that model size and prompting strategy can affect both predictive performance and inference time [3]. Huang and He similarly explored the use of LLM in-context learning for text organization and demonstrated the potential of using LLMs without task-specific embedding fine-tuning [4]. These studies establish the usefulness of LLMs for text classification while also emphasizing the practical trade-off between capability and inference cost [6], [8].

Active Learning addresses the labeling bottleneck by allowing a learner to select informative unlabeled instances for annotation rather than requiring uniform labeling of the entire pool. Uncertainty sampling is a widely studied strategy in which low-confidence or high-uncertainty samples are preferentially selected [1], [2]. The effectiveness of such approaches depends on the quality of the uncertainty signal and on the cost of obtaining additional labels. In practical classification systems, this creates an opportunity to connect uncertainty estimation with resource allocation rather than using uncertainty only as a query criterion [14], [15].

Recent work has also examined the combination of LLM-based annotation and human annotation. Rouzegar studied cost-effective active learning using LLM and human annotators and investigated uncertainty-based mechanisms for deciding when human involvement is required [9]. Longo et al. explored the use of LLM-generated synthetic data to reduce reliance on manual annotation in hierarchical text classification [16]. Other work has investigated LLMs as active annotators and the use of LLMs within active-learning workflows [10]–[13]. These directions demonstrate that LLMs can complement human supervision, but they do not directly address the progressive allocation of classification effort across a lightweight classifier, a local LLM, and human annotation. Our work investigates this three-level resource-allocation setting and evaluates its accuracy–cost behavior across datasets with substantially different classification difficulty.

## III. BACKGROUND

A text classifier produces a probability distribution over the target categories. Let  $p = (p_1, \dots, p_K)$  denote the predicted distribution over  $K$  classes. The uncertainty of a prediction can be quantified using Shannon entropy:

$$H(p) = - \sum_{i=1}^K p_i \log_2 p_i. \quad (1)$$

Maximum class probability,  $c = \max_i p_i$ , provides a complementary confidence measure. Entropy reflects the concentration of the full probability distribution, whereas maximum probability focuses on the most likely class. Using both signals allows the router to identify cases in which the leading class is insufficiently dominant or the overall distribution remains diffuse [1], [15].

Active Learning uses an unlabeled pool to acquire additional labels that are expected to improve a classifier [1], [2]. In this work, the acquisition mechanism is coupled with the routing system: samples reaching the human tier are treated as acquired annotations and added to an accumulated learning history. The classifier is then adapted using this history, allowing the initial model to learn from samples that were sufficiently difficult to trigger human annotation [9], [12].

Confidence calibration is important when uncertainty is used to control downstream actions. A model can be accurate while producing probabilities that do not reflect actual correctness likelihoods [17]. The implementation therefore evaluates Expected Calibration Error (ECE) and Brier score in addition to classification accuracy and macro-F1 [17], [18]. Temperature scaling is fitted on the validation partition before routing thresholds are selected, following the established use of post-hoc temperature scaling for neural-network calibration [17].

## IV. PROPOSED FRAMEWORK

The proposed framework is designed around a resource-allocation principle: the amount of processing applied to an instance should depend on how difficult that instance appears to the current classifier [5], [6], [8]. Instead of applying the local LLM or human annotation uniformly, the system starts with a lightweight encoder and escalates only selected cases. The resulting hierarchy consists of an autonomous encoder tier, an LLM-assisted tier, and a human-annotation tier.

### A. Framework Architecture

For each input instance, Tier 1 produces a class prediction and a calibrated probability distribution. The uncertainty engine evaluates this distribution and decides whether the prediction is sufficiently reliable. Confident instances remain at Tier 1 and incur only the base computational cost. Uncertain instances enter Tier 2, where a local LLM independently reassesses the text using task-specific category definitions and representative examples. Tier 3 is reserved for extreme uncertainty or failed Tier 2 inference.

The architecture is intentionally asymmetric. Tier 2 is not treated as a universal replacement for Tier 1. When Tier 1 and Tier 2 agree, the system records a Tier 2-assisted decision while retaining the shared label. When they disagree, the LLM can override the Tier 1 prediction only when its confidence and vote-agreement conditions satisfy the routing policy. Otherwise, the system falls back to the Tier 1 prediction. This selective override is important because a more capable model is not automatically a more appropriate decision-maker for every uncertain sample [14], [15], [19].

### B. Tier 1: Lightweight Encoder

Tier 1 uses DistilBERT with a task-specific classification head [20]. The pretrained encoder is kept frozen during the active-learning adaptation stage, while the classification components are trained on the target categories. Inputs are tokenized with a maximum sequence length of 128 tokens. The initial classifier is trained on the initial labeled partition before calibration and routing are performed.

The choice of a lightweight encoder provides a low-cost first stage that can process the entire evaluation stream. It also produces a probability distribution required by the uncertainty engine. The model therefore serves both as the primary autonomous classifier and as the source of the routing signal that determines whether additional resources should be used.

### C. Tier 2: Local LLM Assistance

Tier 2 uses Qwen2.5 3B through a local Ollama deployment [21]. The LLM receives dataset-specific category definitions, representative examples, the input text, and the Tier 1 prediction. The prompt is designed to make the second-stage model reassess the initial decision rather than classify the instance without context. Three LLM votes are used in the standard configuration, and vote agreement is retained as an additional confidence signal.

The local deployment is central to the framework’s cost-efficiency objective. It avoids requiring every input to be sent to a remote high-capacity model and allows the intermediate stage to be invoked only when the lightweight classifier is uncertain [5]–[8]. The implementation also supports an offline fallback mode when the local LLM service is unavailable; however, the reported primary configuration is based on the local Qwen2.5 3B tier.

### D. Tier 3: Human Annotation

Tier 3 represents the highest-intervention stage of the framework. It is intended for instances that are extremely uncertain or cannot be resolved reliably by Tier 2. Because the experiments are designed for controlled and repeatable evaluation, human annotation is simulated using a configurable annotation error model rather than measured with human participants. The simulated labels are nevertheless treated as acquired labels for the active-learning process [9], [14].

This distinction is important for interpreting the results. The reported human-effort ratio measures the fraction of test instances routed to the simulated human tier; it does not represent real annotation time, monetary wages, or inter-annotator agreement. The simulation allows the routing policy and resource trade-offs to be compared consistently across datasets while leaving real human evaluation as future work.

### E. Uncertainty-Driven Routing

The router uses two signals derived from Tier 1: entropy  $H(p)$  and maximum confidence  $c$ . Given an entropy threshold  $\tau_1$  and confidence threshold  $\tau_2$ , an instance enters the uncertainty region when

$$H(p) > \tau_1 \quad \text{or} \quad c < \tau_2. \quad (2)$$

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### Algorithm 1 Uncertainty-Driven Tri-Tier Routing

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**Require:** Text  $x$ , Tier 1 probabilities  $p$ , thresholds  $\tau_1, \tau_2$ , extreme cap  $\tau_e$

- 1: Predict Tier 1 label  $y_1$  and confidence  $c = \max(p)$
- 2: Compute entropy  $H(p)$
- 3: **if**  $H(p) > \tau_e$  **then**
- 4:   Escalate  $x$  to Tier 3
- 5: **else if**  $H(p) > \tau_1$  **or**  $c < \tau_2$  **then**
- 6:   Obtain Tier 2 prediction  $y_2$  and vote agreement
- 7:   **if** Tier 2 fails **then**
- 8:     Escalate  $x$  to Tier 3
- 9:   **else if**  $y_1 = y_2$  **then**
- 10:     Return  $y_1$  as Tier 2-assisted result
- 11:   **else if** Tier 2 override conditions are satisfied **then**
- 12:     Return  $y_2$
- 13:   **else**
- 14:     Return  $y_1$
- 15:   **end if**
- 16: **else**
- 17:   Return  $y_1$
- 18: **end if**

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An additional extreme-entropy cap is used to identify cases that should bypass the intermediate stage and proceed directly to Tier 3. For instances in the intermediate uncertainty region, Tier 2 is invoked.

After Tier 2 returns, the router checks whether the LLM prediction agrees with Tier 1. Agreement produces a Tier 2-assisted decision without changing the class. For disagreement, the LLM prediction is accepted only if its vote agreement and confidence conditions satisfy the configured override rule. Otherwise, the Tier 1 prediction is retained. If Tier 2 fails, the instance is escalated to Tier 3. This selective escalation is related to selective prediction and learning-to-defer approaches in which uncertain cases are transferred to a more capable decision-maker [14], [15], [19].

### F. Active Learning Adaptation

The active-learning component uses an unlabeled pool that is disjoint from both the calibration partition and the official test set. When an instance from the active-learning pool reaches Tier 3, its acquired label is added to an accumulated annotation history. The Tier 1 classification components are then adapted using the accumulated history rather than only the most recent sample. This experience-replay design reduces the risk of forgetting earlier acquired examples during sequential updates [1].

The resulting loop can be viewed as a feedback mechanism: uncertainty identifies difficult samples, the human tier provides additional supervision for a selected subset, and the acquired examples are fed back into the lightweight classifier. The direct effect of this loop is evaluated by comparing pre-AL and post-AL Tier 1 accuracy on the untouched test set. The results show that this effect is dataset-dependent rather than uniformly positive in magnitude.

## V. EXPERIMENTAL METHODOLOGY

### A. Datasets and Data Partitioning

Four benchmark datasets are evaluated: AG News, DBpedia-14, Emotion, and IMDB. They cover news categorization, encyclopedic entity categorization, emotion classification, and binary sentiment classification, respectively [22]–[25]. Their different class counts and difficulty profiles provide a useful setting for examining whether the routing policy behaves consistently across tasks.

The implementation enforces four non-overlapping partitions: an initial labeled training set, an active-learning pool, a validation calibration set, and an official test set. The initial training partition contains 2,500 samples for each reported dataset. The active-learning pool contains up to 1,000 samples and the validation partition contains 500 samples. The test set contains 500 samples for AG News, Emotion, and IMDB, and 1,000 samples for DBpedia-14. The official test partition is never used for training, threshold selection, or calibration. The resulting dataset partitions are summarized in Table I.

TABLE I  
DATASETS AND EXPERIMENTAL PARTITIONS

This table summarizes the datasets, classification tasks, class counts, and partitions used for training, active learning, and final testing.

| Dataset    | Task                     | Classes | Initial Train | AL Pool | Test  |
|------------|--------------------------|---------|---------------|---------|-------|
| AG News    | News categorization      | 4       | 2,500         | 1,000   | 500   |
| DBpedia-14 | Entity categorization    | 14      | 2,500         | 1,000   | 1,000 |
| Emotion    | Emotion classification   | 6       | 2,500         | 1,000   | 500   |
| IMDb       | Sentiment classification | 2       | 2,500         | 1,000   | 500   |

*Note*—The active-learning pool and validation partition are drawn from the training split; the test partition is taken from the official test split and remains untouched.

### B. Model and Training Configuration

The Tier 1 model uses `distilbert-base-uncased`, a maximum sequence length of 128, batch size 16, and seven training epochs in the supplied configuration [20]. During active-learning adaptation, the encoder remains frozen and the classification components are updated. The local Tier 2 model is Qwen2.5 3B served through Ollama [21], with three votes in the standard configuration. The Tier 3 human error rate is configured to 5% for simulation. The complete configuration is summarized in Table II.

TABLE II  
CORE MODEL CONFIGURATION

This table lists the principal model, inference, training, and annotation-simulation settings used in the experiments.

| Component               | Configuration                  |
|-------------------------|--------------------------------|
| Tier 1 backbone         | DistilBERT-base-uncased        |
| Input length            | 128 tokens                     |
| Initial batch size      | 16                             |
| Initial training epochs | 7                              |
| Tier 1 adaptation       | Classification components only |
| Tier 2 model            | Qwen2.5 3B                     |
| Tier 2 deployment       | Local Ollama service           |
| Tier 2 votes            | 3                              |
| Tier 3                  | Simulated human annotation     |
| Simulated human error   | 5%                             |

### C. Calibration and Threshold Selection

Before routing, Tier 1 confidence is calibrated using temperature scaling on the validation partition [17]. The temperature parameter is optimized by minimizing validation negative log-likelihood. The calibrated probabilities are then used to compute entropy and maximum confidence.

The routing policy is not selected from the official test set. A validation sweep evaluates candidate entropy and confidence thresholds and selects an operating point using the configured utility criteria. The implementation searches entropy values of 0.3, 0.5, 0.7, 0.8, and 0.9 together with confidence values of 0.4, 0.5, 0.6, and 0.7, while retaining dataset-specific configured values when they are outside the grid. The extreme-entropy cap is calibrated from a validation percentile. The selected routing policy is then fixed before the primary test evaluation. The resulting dataset-specific thresholds are reported in Table III.

TABLE III  
VALIDATION-SELECTED ROUTING THRESHOLDS USED IN THE PRIMARY EXPERIMENTS

This table reports the entropy and confidence thresholds selected from validation data for each dataset.

| Dataset    | $\tau_1$ (Entropy) | $\tau_2$ (Confidence) | Selection Basis  |
|------------|--------------------|-----------------------|------------------|
| AG News    | 1.00               | 0.40                  | Validation sweep |
| DBpedia-14 | 0.30               | 0.40                  | Validation sweep |
| Emotion    | 0.30               | 0.60                  | Validation sweep |
| IMDb       | 0.90               | 0.50                  | Validation sweep |

### D. Baselines and Ablations

The primary system is compared with a Tier-2-only baseline, a no-Tier-2 ablation, a no-entropy routing ablation, and a budget-matched random routing baseline. The Tier-2-only baseline routes all samples to the local LLM and therefore tests whether the proposed hierarchy provides value over unconditional LLM use. The no-Tier-2 ablation bypasses the intermediate model and directly escalates uncertain samples to Tier 3, providing an upper-human-effort comparison. The no-entropy ablation removes the entropy signal while retaining confidence-based routing. The random-routing baseline uses a comparable human-effort budget to test whether gains arise from informed uncertainty-based selection rather than escalation alone.

Additional Active Learning baselines are implemented for random selection, least-confidence selection, entropy selection, and margin-based selection. These baselines provide alternative acquisition policies for the same active-learning setting and help distinguish the proposed tri-tier routing mechanism from standard query strategies [1], [2].

### E. Evaluation Metrics

Classification quality is measured using accuracy and macro-F1. Calibration is assessed using ECE and the multiclass Brier score [17], [18]. Resource usage is characterized using the human-effort ratio and normalized compute cost. If  $N_1$ ,  $N_2$ , and  $N_3$  denote the number of samples processed at the three tiers, the average compute cost is

$$C_{avg} = \frac{N_1 c_1 + N_2 c_2 + N_3 c_3}{N}, \quad (3)$$

where the configured relative costs are  $c_1 = 1$ ,  $c_2 = 200$ , and  $c_3 = 3000$ .

The human-effort ratio is defined as  $r_h = N_3/N$ . The Point-Improvement-per-Cost-Ratio (PICR) used by the evaluator is

$$\text{PICR} = \frac{\Delta A}{r_h}, \quad (4)$$

where  $\Delta A$  is the final accuracy improvement over Tier 1. A complementary active-learning-aware form incorporates the pre-AL to post-AL Tier 1 improvement. The human-effort utility is computed as

$$U_h = \Delta A - \lambda_h r_h, \quad (5)$$

with  $\lambda_h = 0.05$ . A cost-aware utility similarly penalizes normalized compute cost. These measures are used as comparative indicators rather than as monetary estimates.

### F. Reproducibility Protocol

The final experiments are executed with multiple random seeds and retain all valid runs, including runs with zero human escalation. The supplied implementation records per-dataset metrics, confusion matrices, per-class precision/recall/F1, routing distributions, threshold sweeps, active-learning learning curves, and multi-seed summaries. Automated tests cover dataset partition non-overlap, routing modes, calibration and metric computations. This protocol is intended to reduce accidental test leakage and make the reported trade-offs reproducible.

## VI. RESULTS AND DISCUSSION

### A. Primary Test-Set Performance

Table IV summarizes the primary test-set evaluation. The proposed framework improves final accuracy over the post-AL Tier 1 classifier on every dataset. The absolute gains are 1.60 percentage points on AG News, 0.40 points on DBpedia-14, 2.60 points on Emotion, and 8.00 points on IMDB. Table IV reports the corresponding macro-F1, human-effort, and average-cost values. The largest gain occurs on IMDB, where the lightweight classifier leaves substantial room for improvement and 45.0% of instances enter Tier 2 while 3.6% reach Tier 3. DBpedia-14 shows the smallest gain because its Tier 1 accuracy is already 97.90%. The cross-dataset synthesis is shown in Fig. 1.

TABLE IV

PRIMARY TEST-SET RESULTS OF THE PROPOSED FRAMEWORK

This table compares the post-active-learning Tier 1 classifier with the routed system on accuracy, macro-F1, human effort, and average compute cost.

| Dataset    | T1 Acc. | Final Acc. | $\Delta$ Acc. | Macro-F1 | Human Effort | Cost (Avg.) |
|------------|---------|------------|---------------|----------|--------------|-------------|
| AG News    | 87.60%  | 89.20%     | +1.60 pp      | 0.8931   | 0.40%        | 46.43       |
| DBpedia-14 | 97.90%  | 98.30%     | +0.40 pp      | 0.9835   | 0.20%        | 20.13       |
| Emotion    | 60.20%  | 62.80%     | +2.60 pp      | 0.3994   | 1.00%        | 226.81      |
| IMDb       | 79.40%  | 87.40%     | +8.00 pp      | 0.8734   | 3.60%        | 198.51      |

Note—Average cost is reported in normalized compute-cost units from the supplied evaluator.

The cross-dataset synthesis is shown in Fig. 1.

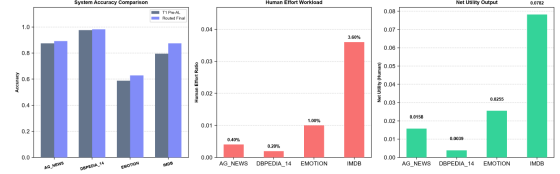


Fig. 1. Cross-dataset synthesis from the supplied research report: primary system accuracy, simulated human effort, and human-effort net utility.

This figure compares final accuracy, simulated human effort, and net utility across the four datasets.

### B. Routing Distribution and Resource Allocation

The routing distributions and accuracy waterfalls are shown in Fig. 2.

The routing distribution provides a direct view of the framework’s operating point. Across the primary runs, Tier 1 handles 82.8% of AG News samples, 96.0% of DBpedia-14, 0.8% of Emotion, and 51.4% of IMDB. The remaining instances are primarily processed by Tier 2, with only a small fraction reaching Tier 3. The routing and waterfall panels in Fig. 2 show that the local LLM is most heavily used on datasets for which the lightweight classifier is less decisive [5], [6], [8].

The human-effort results further illustrate this trade-off. DBpedia-14 uses only 0.20% simulated human effort, while IMDB uses 3.60%. The normalized cost model produces positive cost-aware net utility on all four datasets, with values from 0.0037 on DBpedia-14 to 0.0767 on IMDB. The same architecture therefore operates at different points on the accuracy–resource frontier depending on the uncertainty profile of the underlying classifier [5], [6].

### C. Per-Class Behavior and Active Learning

The per-class results indicate that the gains are not uniformly distributed across categories. In AG News, final F1 improves over Tier 1 for World, Sports, Business, and Sci/Tech. The report also shows that escalation is concentrated in classes for which the initial classifier is less decisive, such as Business and Sci/Tech. Similar dataset-specific patterns appear in DBpedia-14, Emotion, and IMDB. This supports the interpretation of routing as instance- and class-sensitive rather than as a uniform post-processing step [1], [15].

The active-learning loop itself produces smaller and more variable gains than the full routing system. Pre-AL to post-AL Tier 1 accuracy changes from 87.40% to 87.60% on AG News, 97.60% to 97.90% on DBpedia-14, 58.80% to 60.20% on Emotion, and remains 79.40% on IMDB. Thus, the largest system-level improvements should not be attributed solely to classifier adaptation; they also arise from the Tier 2 and Tier 3 decisions made during routing [9], [12].

### D. Multi-Seed Robustness

The accuracy–human-effort frontier and active-learning tracking are shown in Fig. 3.

The five-seed evaluation provides a robustness check beyond the single primary run. Table V reports the corresponding means, confidence intervals, human-effort ratios, and final ECE values. Mean final accuracy is 88.56% on AG News, 98.76%

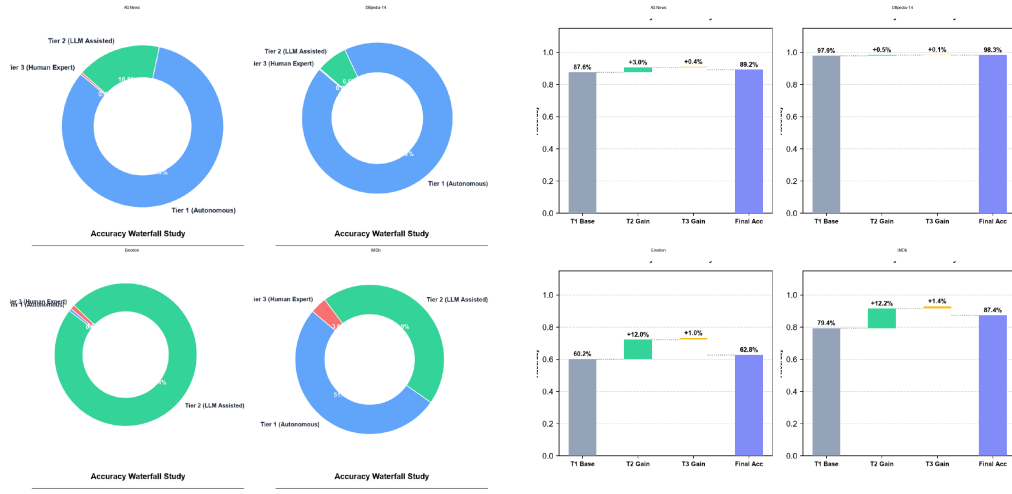


Fig. 2. Request-routing distributions and accuracy waterfalls for the four primary datasets, reproduced from the supplied evaluation report. This figure shows the distribution of requests across the three tiers and the corresponding accuracy contribution of each tier.

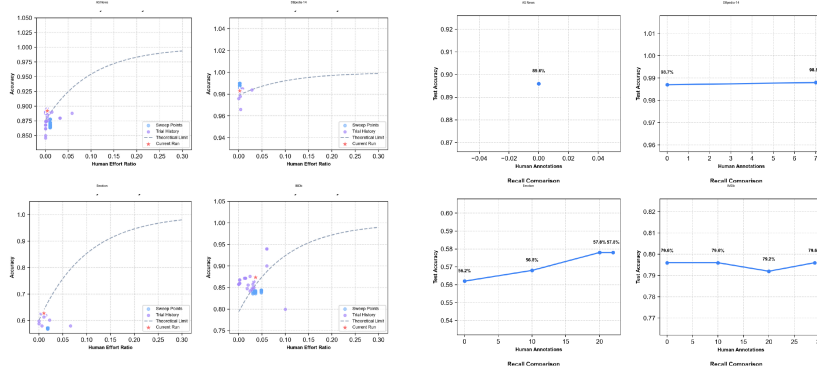


Fig. 3. Accuracy-human-effort frontier and active-learning tracking across the four datasets.

This figure shows the accuracy-human-effort frontier and the change in Tier 1 accuracy as additional annotations are acquired.

on DBpedia-14, 59.44% on Emotion, and 86.00% on IMDb. Relative to the corresponding mean Tier 1 accuracies, the mean gains are +0.28, +0.26, +1.68, and +5.80 percentage points, respectively.

TABLE V  
FIVE-SEED ROBUSTNESS SUMMARY (MEAN WITH 95% CONFIDENCE INTERVAL)

This table summarizes the mean performance, improvement, human effort, and calibration error across five random seeds.

| Dataset    | T1 Mean           | Final Mean        | Gain     | Human Effort     | Final ECE |
|------------|-------------------|-------------------|----------|------------------|-----------|
| AG News    | 88.28% $\pm$ 0.86 | 88.56% $\pm$ 0.71 | +0.28 pp | 0.56% $\pm$ 0.50 | 0.0465    |
| DBpedia-14 | 98.50% $\pm$ 0.30 | 98.76% $\pm$ 0.24 | +0.26 pp | 0.36% $\pm$ 0.23 | 0.0062    |
| Emotion    | 57.76% $\pm$ 2.93 | 59.44% $\pm$ 2.33 | +1.68 pp | 1.64% $\pm$ 0.98 | 0.2141    |
| IMDb       | 80.20% $\pm$ 2.09 | 86.00% $\pm$ 2.06 | +5.80 pp | 2.48% $\pm$ 1.03 | 0.0772    |

Note—The reported intervals are 95% confidence intervals from the supplied five-seed summaries.

The direction of improvement remains positive for all four datasets, although the uncertainty is larger on Emotion and IMDb. The seed-level variation also shows that the routing policy is sensitive to the initial classifier and the selected operating point. This supports reporting the multi-seed results alongside the primary run rather than relying on a single random

seed.

### E. Baseline and Ablation Analysis

The baseline and ablation comparison is shown in Fig. 4.

The baseline and ablation results clarify the purpose of the intermediate LLM tier. The no-Tier-2 configuration produces higher accuracy than the proposed system on all four datasets, but it does so by allowing much greater human involvement. In the primary runs, human effort rises to 16.80% on AG News, 2.50% on DBpedia-14, 94.00% on Emotion, and 65.20% on IMDb in the no-Tier-2 setting. This demonstrates that additional human supervision can provide substantial accuracy, but it also shows why maximizing accuracy alone does not capture the objective of the proposed framework [14], [19].

The Tier-2-only baseline gives 86.40% on AG News and 96.30% on DBpedia-14, below the corresponding Tier 1 accuracies, while reaching 61.00% on Emotion and 89.20% on IMDb. These results show that the local LLM is not uniformly better than the lightweight classifier. The proposed router therefore treats Tier 2 as selective assistance rather than as a universal replacement [5], [6], [8].

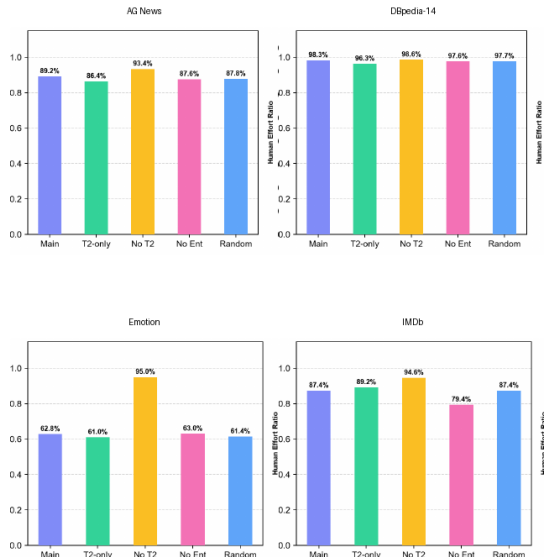


Fig. 4. Accuracy comparison across the proposed system and primary baselines/ablations for the four datasets.

This figure compares the proposed framework with the principal baselines and routing ablations across the four datasets.

TABLE VI  
PRIMARY BASELINE AND ABLATION ACCURACIES

This table compares the proposed routing policy with the main baselines and routing ablations across all datasets.

| Dataset    | Proposed | Tier-2 Only | No Tier-2 | No Entropy | Random |
|------------|----------|-------------|-----------|------------|--------|
| AG News    | 89.20%   | 86.40%      | 93.40%    | 87.60%     | 87.80% |
| DBpedia-14 | 98.30%   | 96.30%      | 98.60%    | 97.60%     | 97.70% |
| Emotion    | 62.80%   | 61.00%      | 95.00%    | 63.00%     | 61.40% |
| IMDb       | 87.40%   | 89.20%      | 94.60%    | 79.40%     | 87.40% |

*Note*—The no-Tier-2 configuration intentionally permits substantially greater human intervention.

The no-entropy ablation removes one of the two uncertainty signals used by the standard router. Its weaker escalation behavior on several datasets supports the use of entropy as a routing signal. The random-routing baseline provides a complementary check: if similar accuracy could be obtained through arbitrary escalation under the same budget, the uncertainty mechanism would have limited value. The observed differences indicate that the routing policy contributes to the allocation of effort, although the degree of benefit remains dataset-dependent [1], [15].

#### F. Calibration and Confusion-Matrix Analysis

The calibration diagrams and final confusion matrices are shown in Fig. 5.

Calibration behavior varies substantially across datasets. The final ECE values are 0.0564 for AG News, 0.0143 for DBpedia-14, 0.3067 for Emotion, and 0.0719 for IMDb. DBpedia-14 therefore has the most reliable final confidence estimates among the four primary runs, while Emotion exhibits the largest calibration error. This difference is consistent with the more difficult six-class setting and the much higher reliance on Tier 2 for Emotion [17].

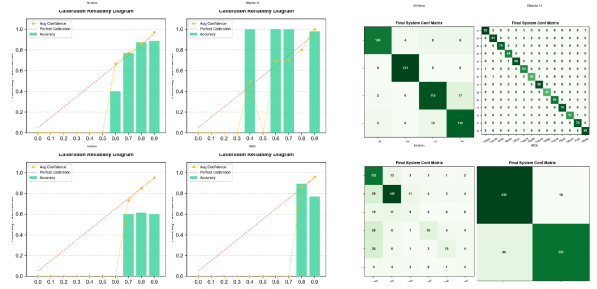


Fig. 5. Calibration reliability diagrams and final-system confusion matrices for the four datasets.

This figure presents confidence calibration and final-system confusion matrices for the evaluated datasets.

The final confusion matrices in Fig. 5 show that routing changes the distribution of errors rather than simply increasing every class equally. For AG News, for example, the final system reduces several off-diagonal errors involving World, Business, and Sci/Tech while maintaining strong Sports performance. IMDb shows a more pronounced shift in the confusion matrix, consistent with its large eight-point accuracy gain.

#### G. Cost and Utility Interpretation

The evaluator assigns relative costs of 1, 200, and 3000 units to Tier 1, Tier 2, and Tier 3, respectively. Under this model, the average costs range from 20.13 for DBpedia-14 to 226.81 for Emotion. The human-effort utility is positive for all primary runs, with values of 0.0158, 0.0039, 0.0255, and 0.0782 for AG News, DBpedia-14, Emotion, and IMDb, respectively. The cost-aware utility is also positive, ranging from 0.0037 to 0.0767.

These utility values should be interpreted comparatively rather than as economic forecasts. They indicate whether the observed accuracy improvement outweighs the configured penalty under the experimental cost model. The cross-dataset synthesis in Fig. 1 shows the same pattern visually: IMDb obtains the largest accuracy and utility improvement but also requires the greatest human effort among the four primary runs.

#### VII. LIMITATIONS AND THREATS TO VALIDITY

Several limitations should be considered when interpreting the results. First, the human annotation tier is simulated with a configurable error model rather than evaluated with real human annotators. Consequently, the reported human-effort and utility results represent controlled simulation rather than measured annotation time, annotator disagreement, or expert reliability. A real human study may produce a different accuracy–effort relationship [9], [14].

Second, the local LLM tier depends on Qwen2.5 3B and dataset-specific prompts [21]. Different local models, quantization settings, prompts, or inference temperatures may produce different routing outcomes. The framework therefore demonstrates a particular local-LLM configuration rather than establishing that all local LLMs provide the same intermediate-tier behavior [8], [12].

Third, the normalized cost values are relative weights rather than measured monetary or energy costs. They are useful for comparing routing strategies under a common evaluator, but they should not be interpreted as deployment prices. The cost model also assigns a fixed relative cost to each tier, whereas real deployments would have hardware-dependent latency and energy characteristics [5], [6].

Fourth, the primary experiments evaluate 500 test samples for three datasets and 1,000 for DBpedia-14. Although the supplied implementation includes multiple seeds and several baselines, broader datasets and larger evaluation pools would provide stronger evidence about scalability. Finally, the direct improvement of Tier 1 through active learning is modest on some datasets and zero in the primary IMDb run. The strongest evidence therefore concerns the complete routing system rather than a universal claim that the active-learning loop alone improves the lightweight classifier [12], [26].

## VIII. CONCLUSION AND FUTURE WORK

This paper presented a tri-tiered local LLM framework for cost-efficient active learning in text classification and automated categorization. The framework combines a lightweight DistilBERT classifier [20], local Qwen2.5 3B inference [21], and a simulated human annotation tier through uncertainty-driven routing. Entropy and confidence are used to decide when additional processing is warranted, while validation-based calibration and threshold selection preserve the isolation of the official test set [17]. Human-tier labels are accumulated for active-learning adaptation of the lightweight classifier [9].

Across AG News, DBpedia-14, Emotion, and IMDb, the primary test runs improved final accuracy over Tier 1 by 0.4–8.0 percentage points while limiting simulated human effort to 0.2–3.6%. Five-seed summaries retained positive mean accuracy gains across all four datasets. The ablation results further show that substantially higher accuracy can be obtained by allowing much greater human involvement, confirming that the central objective is not accuracy alone but the allocation of accuracy gains against computational and annotation resources [14], [19].

Future work should replace simulated human annotation with real annotation studies, evaluate additional local LLMs and quantization levels, investigate learned or adaptive routing policies, and extend the cost model using measured inference latency, energy, and annotation time. Larger and more diverse datasets should also be used to test whether the uncertainty-driven hierarchy remains effective across domains. Another promising direction is to jointly learn routing thresholds and the Tier 1 classifier so that the system can adapt its operating point as the active-learning pool evolves [8], [13].

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